Problem 2 · Developer Community Pulse — Components Map

Sumit's code-review startup · weekly brief every Friday morning

Actor	Trigger	Input	Context	Processing	AI	Output
Sumit Product Manager	Clicks "Run now" on the dashboard for an on-demand brief	Adds or edits topic categories and keywords on the /topics page, with no code changes				Reads the brief before the afternoon sync and plans Monday's sprint with data
Scheduler Vercel Cron	Fires every Friday 03:00 UTC (8:30 AM IST) and calls /api/run-digest					
Dev Communities HN · Stack Overflow · GitHub · Dev.to		Live posts from the last 7 days matching each keyword, with score, comments, answered?				
Database Supabase			topics table (categories + keywords) and last week's metrics and pain-point labels	SQL counts mentions per pain point, compares with last week: Rising / Falling / New		Saves posts, weekly metrics and the digest, building the history for next week
Automation Next.js app on Vercel				Queries each source per keyword, removes duplicates, sends posts to Gemini in batches		
AI Model Gemini Flash				Gemini tags each post: category, pain point, sentiment, unresolved or not	Gemini writes the brief: top pain points, trends, unsolved gaps, sales objections, 3 actions	
Email Resend						Friday-morning brief in Sumit's inbox, every claim linked to a real thread
Dashboard Web app on Vercel						Latest brief, pain-point trend table and chart, past briefs, Run now button

The cut: AI tags and writes, code counts

Gemini only labels posts and writes the brief. Mention counts and the week-over-week % change are plain SQL, so "false positive rate: 4 → 14" is exact and repeatable.

Why stable pain-point labels

Gemini is given last week's labels from Supabase and reuses them, so "false positives" and "too many false alarms" land in one bucket and trends stay comparable.

Running cost

Estimated ~300 posts a week ≈ 100k Gemini Flash tokens: inside the free tier, a few rupees a month if paid. HN, Stack Exchange, GitHub, Supabase, Resend and Vercel Hobby are free.

Not built now: Reddit (needs OAuth from Vercel servers) and Slack delivery plug in at the same Input and Output steps, and everything in between stays the same.